23.1 Ancestry and Features of Land Plants

Learning Outcomes

Upon completion of this section, you should be able to

1. Compare and contrast algae with land plants.
2. List the traits that enabled plants to adapt to life on land.
3. Evaluate the differences in the alternation of generations of land plants.

Plants are multicellular, photosynthetic eukaryotes whose evolution is marked by adaptations to a land existence. Plant systematists use molecular and morphological information to classify plants. Plants are in the supergroup Archaeplastida, along with the red algae, green algae, and charophytes (see Fig. 21.1). In our discussions, the plants will be described as nonvascular (bryophytes), seedless vascular (lycophytes, ferns, and fern allies), and vascular seed plants (gymnosperms and angiosperms). Table 23.1 provides a summary.

Phylogenetic information suggests that the modern plant groups arose in a particular sequence (Fig. 23.1). Liverworts diverged first, mosses diverged next, and hornworts seem to be closely related to vascular plants. Vascular plants are distinguished by tissues that conduct food and water and provide structural support. Lycophytes are then related to all the other modern vascular plants; ferns arose next, then the seed plants.

Plants are thought to have evolved from a freshwater green algae. In this section, we consider similarities between green algae and plants, as well as the different adaptations plants have to facilitate life on land.

Table 23.1 Characteristics of Plants

DOMAIN: Eukarya
SUPERGROUP: Archaeplastida
KINGDOM: Plantae

CHARACTERISTICS

Multicellular; apical tissue produces specialized tissues; alternation-of-generations life cycle; protection of the embryo; gametangia produces gametes; waxy cuticle prevents water loss.

Nonvascular Plants—The Bryophytes

Liverworts, hornworts, and mosses are low-lying and live in moist environments. Dominant gametophyte produces flagellated sperm; unbranched, dependent sporophyte produces windblown spores.

Seedless Vascular Plants—The Lycophytes and Pteridiophytes

Dominant; branched sporophyte has vascular tissue: lignified xylem transports water, and phloem transports organic nutrients; typically has roots, stems, and leaves; sporangia borne on sides of leaves that produce windblown spores; independent and separate gametophyte produces flagellated sperm.

- Lycophyte leaves are microphylls with a single, unbranched vein.
- Pteridiophyte leaves are megaphylls with branched veins.

Seeded Vascular Plants—The Gymnosperms and Angiosperms

Leaves are megaphylls; dominant sporophyte produces heterospores that become dependent male and female gametophytes. Male gametophyte is pollen grain, and female gametophyte occurs within ovule, which becomes a seed.

- Gymnosperms are cone-bearing; sporophyte bears pollen cones, which produce windblown pollen (male gametopyte), and seed cones, which produce seeds.
- Angiosperm sporophytes bear flowers, which produce pollen grains, and bear ovules within an ovary. Following double fertilization, ovules become seeds that enclose a sporophyte embryo and endosperm. Fruit develops from ovary.

The Ancestry of Plants

The evolutionary history of plants begins in the water. Most likely, land plants evolved from a form of freshwater green algae. Green algae are not plants, but like plants they (1) contain chlorophylls *a* and *b* and various accessory pigments, (2) store excess carbohydrates as starch, and (3) have cellulose in their cell walls. One major difference, however, between algae and terrestrial plants is that plants not only protect the zygote but also protect and nourish the resulting embryo—an important feature that separates land plants from green algae (Fig. 23.2).

The commonality with green algae, as well as a comparison of DNA and RNA base sequences, suggests that land plants are most closely related to a particular group of freshwater green algae known as **charophytes.** Molecular data show that charophytes and land plants are in the same clade and form a monophyletic group. Although *Spirogyra* (see Fig. 21.21) is a charophyte, the ancestors of land plants were more closely related to the charophytes *Chara* and *Coleochaete,* shown in Figure 23.3.

Chara are commonly known as stoneworts, because they are encrusted with calcium carbonate deposits. The body consists of a single file of very long cells anchored in mud by thin filaments. Whorls of branches occur at regions called *nodes,* located between the cells of the main axis. Male and female reproductive structures grow at the nodes. *Coleochaete* appears disklike, but the body is actually composed of long, branched filaments of cells. Most important to the evolution of plants, charophytes protect the zygote.

Adaptation to Land

Recall what it is like swimming in a pool. Like algae, you are weightless, surrounded by water and protected from the sun. However, all that changes when you leave the water. When plants moved from water to land, they needed mechanisms to deal with water loss, gravity, and sun exposure. The following features are evolutionary adaptions of most terrestrial plants.

- Unlike algae, plants living on land have a limited amount of water available to them. As an adaptation to living on land, most plants are protected from desiccation (drying out) by a waxy covering called a **cuticle** (Fig. 23.4*a*). The cuticle covers all exposed surfaces and is relatively impermeable. The problem is that this limits gas exchange for photosynthesis and cellular respiration. The solution is tiny openings, found mainly on the underside of leaves, called **stomata** (*sing.,* stoma) (Fig. 23.4*b*). These openings, or pores, allow for gas exchange but can also allow water vapor to escape.

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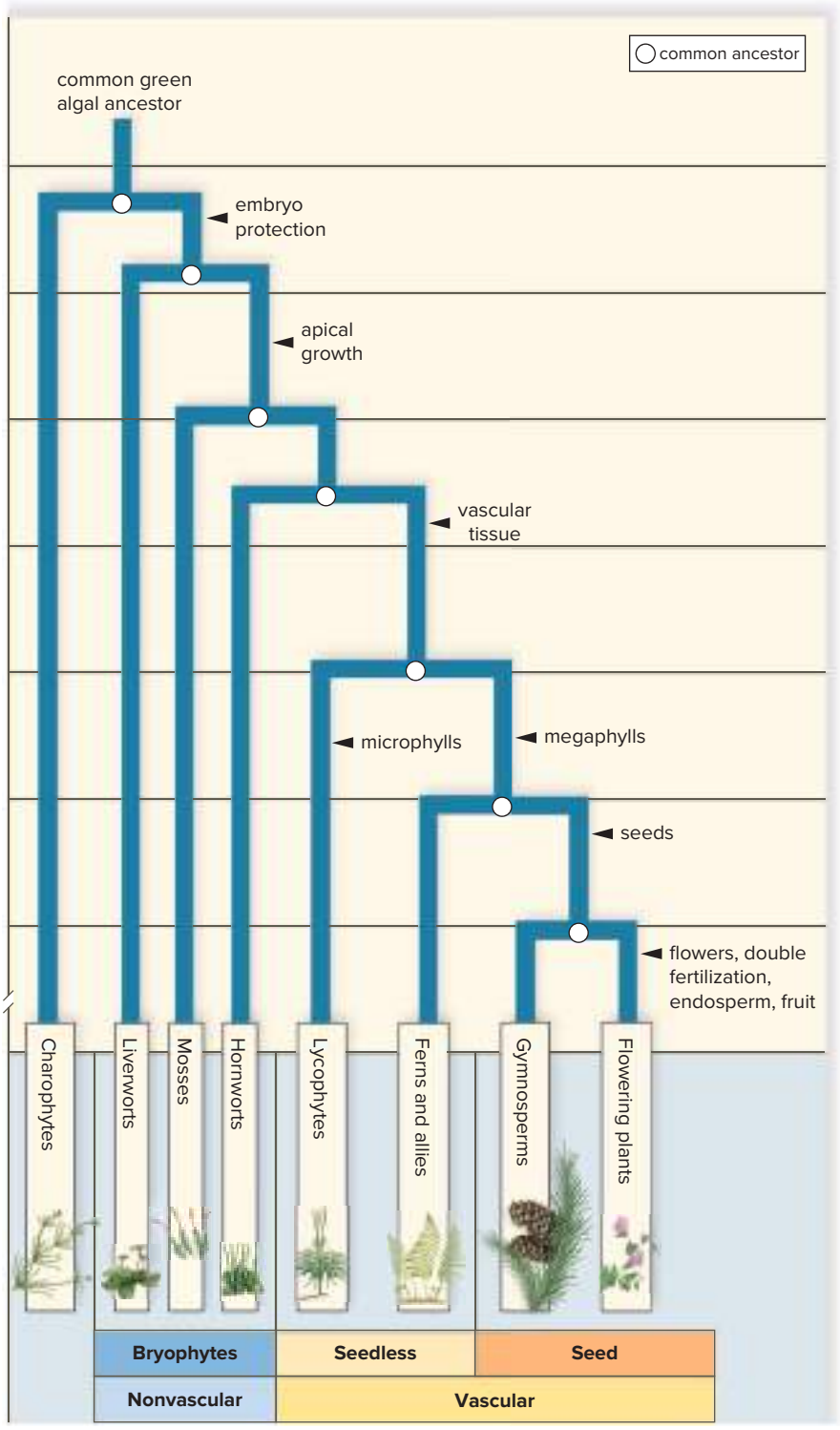


Figure 23.1 Evolutionary history of plants. The evolution of plants involves several significant innovations.

In Section 26.1, we will explore how stomata can be closed at times to limit water loss.

- Moving water within plants is a challenge that increases with plant size. Picture the height of moss compared to the height of a redwood tree. The different members of the land plants can be distinguished based on the presence or absence of **tracheids,** specialized cells with proteins that resist gravity and facilitate the upward transport of water and minerals.
- Compared to green algae, the bodies of land plants are mostly composed of three-dimensional tissues. A *tissue* is an association of many cells of the same type that can lead to specialized structures, such as organs. Tissues provide land plants with an increased ability to avoid water loss at their surfaces because bodies composed of tissues have a lower surface-area-to-volume ratio than do branched filaments, as seen in *Chara* (see Fig. 23.3).
- Plants living on land are exposed to higher intensities of UV rays than are aquatic algae. Exposure to ultraviolet light can increase the chance of mutations, but having a diploid genome can hide the effect of a single, deleterious allele. All terrestrial plants have both a haploid and a diploid generation, and the evolutionary shift toward a more dominant diploid generation allows for greater genetic variability in land plants.

Alternation of Generations

In the life cycle of humans, and other animals, the diploid stage is multicellular, whereas the haploid stage is composed of single-celled gametes (see Fig. 10.9). Plants also have a diploid and a haploid stage, but the significant difference is that both of these stages contain multicellular structures. Therefore, all land plants exhibit an **alternation of generations,** meaning that the plant has two alternating forms in the course of its life cycle. The two types of multicellular bodies are (1) the diploid, spore-producing **sporophyte** generation and (2) the haploid, gamete-producing **gametophyte** generation (Fig. 23.5).

The sporophyte produces a specialized structure called a **sporangium,** where meiosis takes place to produce haploid spores. A **spore** is a reproductive cell that develops into a new organism without the need to fuse with another reproductive cell. The spore then undergoes mitosis and becomes a new multicellular gametophyte.

The gametophyte is a multicellular haploid structure that will develop both male and female gamete-producing regions called *gametangia.* Best seen in the moss life cycle in Figure 23.9, male gametangia are called **antheridia;** they produce sperm. The female gametangia are called **archegonia;** they produce eggs. Sperm and egg will fuse, become a *zygote,* and begin the diploid part of the life cycle.

Dominant Generation

Land plants differ as to which generation is dominant—that is, more conspicuous. In plants such as mosses, the gametophyte is